Ian Leung

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Education

University of Waterloo — **Expected 2029**
Bachelor of Computer Science, Finance, & Statistics – Triple Major — GPA: 3.8/4.0

- President's Scholarship of Distinction (**$5000**), Undergraduate Achievement Award (**$1500**)
- Relevant Coursework: Object-Oriented Programming (Advanced), Sequential Programming (Advanced), Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Statistics, Proofs, Financial Markets & Data Analytics

Experience

Roblox — **Jun 2026 – Aug 2026**
Software Engineer Intern (Network Reliability) | Go, Python — San Mateo, CA

- Automated the network engineering deployment process, pushing to **5000+** production devices in global datacenters
- Built a GitHub Actions CI/CD pipeline, saving **400+ engineering hours** annually by eliminating manual tasks
- Orchestrated Temporal Workflows with LLM analysis & Slack alerting for safe, autonomous deployment of network changes, preventing outages/misconfigurations and identifying issues on **20+** network devices
- Identified and fixed a resource leak in an internal network service, recovering **7%** CPU utilization across devices

Waterloo Quant Club — **Jan 2026 – Present**
Software Engineer (DevOps) | Rust, Python, Docker — Waterloo, ON

- Launched a Grafana observability platform with **100+** metrics, including membership and infrastructure monitoring
- Harnessed Alertmanager to send automatic Discord alerts for critical service failures, maintaining a $>$**99.5%** uptime
- Engineered portal.waterlooquantclub.com, a full-stack club recruitment & event portal hosted on AWS for **700+** users
- Built and deployed 2v2bluff, a real-time multiplayer **Rust/WebSocket** card game supporting four concurrent players

Sunnybrook Research Institute — **Jul 2025 – Aug 2025**
Software Research Intern | Python — Toronto, ON

- Built an event tracing, playback, and logging tool that records and simulates clinician workflows in **Focused Ultrasound** software, surfacing **5+** defects and edge cases in real usage patterns
- Automated regression testing by generating synthetic JSON event flows and validating results against recorded baselines, cutting testing time by **91%**

Projects

Pharmagotchi – Pharmacy Companion App | Kotlin, Android Studio, Figma, OpenRouter

- Built a Tamagotchi-inspired Android companion app to track health & medications and connect users with physicians
- Integrated an **OpenRouter**-powered chat assistant with Retrieval-Augmented Generation (RAG) that parses health conditions and medications into structured JSON vitals and provides context-aware medication guidance
- Connected Fitbit tracking devices through the Google Health API, centralizing health data for seamless monitoring

PeerAssist – Peer Review Platform | Java, Supabase, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for **20+** students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with Apache PDFBox and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

N-body Simulation | Python, Pyglet, NumPy

- Built a 2D particle gravity simulation with tunable constants, supporting **1000+** concurrent particles
- Implemented a Barnes-Hut quadtree algorithm, cutting unnecessary calculations and speeding up simulation by **220+%**
- Designed a graphics renderer to display particles and quadtree bounding boxes using Pyglet, an OpenGL binding

Skills

Languages	Python, Java, C++, C#, Go, Kotlin, TypeScript, SQL, Bash
Technologies	Temporal, gRPC, REST, PyTorch, Pandas, Numpy, OpenCV, PostgreSQL, React, Node.js
Tools	Git, Docker, AWS, Github Actions, Grafana, OpenRouter, Supabase, Android Studio